



BEASTLY FACTS · CARE PACKAGE

RUSSIAN TORTOISE · *AGRIONEMYS HORSFIELDII*

Keep your Russian tortoise the right way from day one.

A complete, checked-fact printable manual: setup, UVB, diet, the brumation decision, and the routines that keep a tortoise thriving for decades.

21 pages

Intermediate

Print or tablet

Version 1.0 · August 2026



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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't set up yet

Read pages 4 through 9 in order before buying anything. Housing, temperature, and UVB decisions are expensive to redo, and the brumation decision on page 7 is easier to plan for before your tortoise's first winter than after. Get the enclosure size and lighting right the first time, then use the Budget & Shopping List on page 15 to buy once.

If your tortoise is already set up

Jump to the Setup Checklist and Temperature/UVB Targets on page 13 to audit what you have, and the Health Red Flags section starting on page 10 if anything seems off.

Print these first

- | | |
|---|---|
| ☐ Page 13: Setup Checklist & Temperature/UVB Targets | ☐ Page 18: Daily, Weekly, Monthly & Seasonal Routine |
| ☐ Page 14: Emergency & Quick Targets Card | ☐ Page 19: Owner Log |
| ☐ Page 17: Symptom Quick Reference | ☐ Page 20: Enrichment Checklist |

WHY THIS PACKAGE EXISTS

Russian tortoise advice online varies wildly, especially around brumation and UVB strength. This package reflects current UVB, enclosure size, and diet standards, cross checked against veterinary and herpetological sources, not recycled forum advice.

NOT A SUBSTITUTE FOR

A reptile experienced veterinarian. This is a husbandry reference for day to day care. Anything urgent or health related is always a vet call, not a page in this book.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Agrionemys horsfieldii</i> (also called <i>Testudo horsfieldii</i>)
Adult size	5 to 10 in (13 to 25 cm), measured along the carapace
Captive lifespan	Commonly 40 years or more with proper care; some individuals reach 50 or beyond
Native range	Steppe and semi-desert of Central Asia: Kazakhstan, Uzbekistan, Iran, Pakistan, China
Diet type	Herbivore; weeds and leafy greens dominate, fruit under 5% of intake if given at all
Enclosure minimum (adult)	4×2×2 ft indoor open-top tortoise table; preferred size closer to 7×3.5 ft
Handling difficulty	Intermediate; comes with a genuine winter brumation decision to manage
Conservation status	Vulnerable (IUCN)

Cost overview

Initial setup

\$400 to \$600 for a complete indoor setup: table, UVB fixture, heat, substrate, and furnishings. The open-topped tortoise table itself runs \$125 to \$200 of that.

Annual ongoing

\$500 to \$650 across the year for substrate refreshes, forage, supplements, and UVB bulb replacement.

BUDGET FOR THE WORST CASE TOO

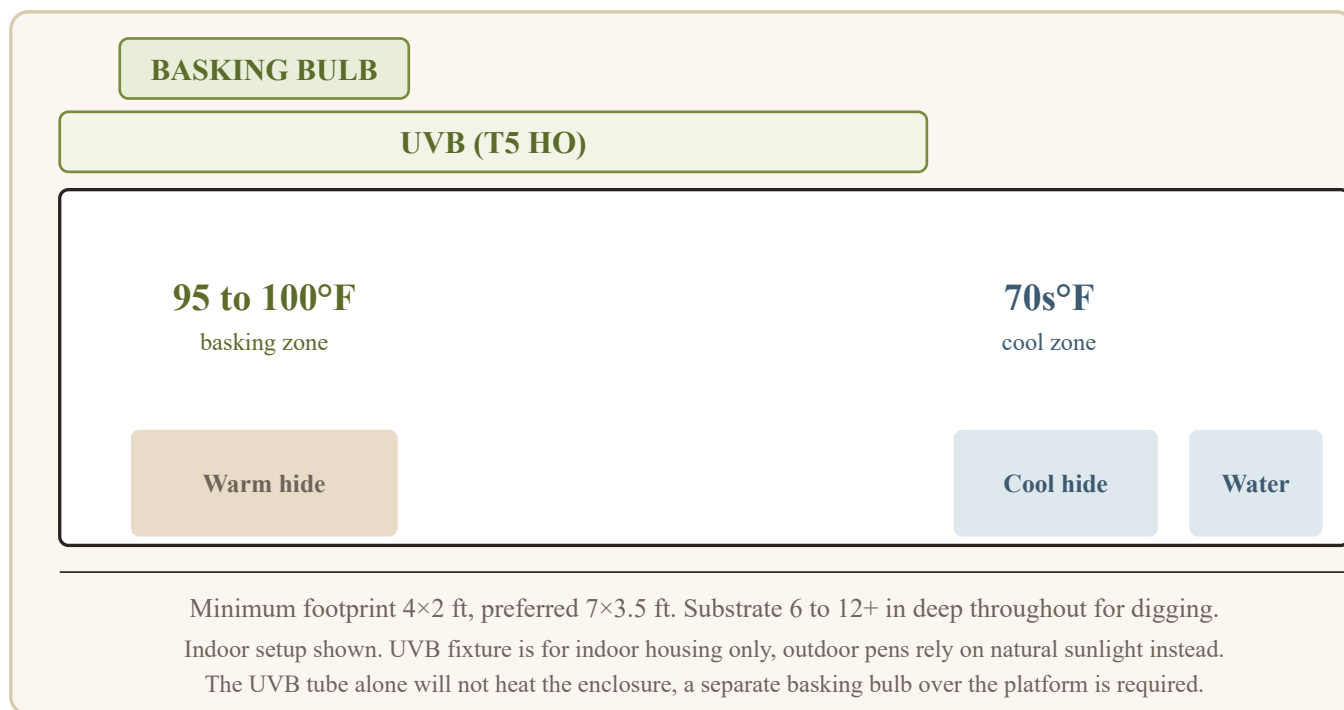
A routine vet checkup runs around \$75; a workup for an illness commonly runs \$150 to \$300 or more. Neither is likely if you follow this package, but both are real enough to plan for.

Fun facts worth knowing

- Russian tortoises are serious diggers, capable of excavating burrows to escape both summer heat and winter cold, which is exactly why outdoor pen walls need to run at least 12 inches below grade.
- They're one of the few pet tortoise species that genuinely offer their keeper a choice between brumating for winter or staying active year-round indoors, a flexibility most other species don't have built in as cleanly.
- Wild collection for the pet trade is part of why this species is listed Vulnerable, which is one more reason captive-bred sourcing matters.

Housing & enclosure

4×2×2 ft is the indoor minimum for an adult, built as an open-top tortoise table rather than a glass tank. Floor space is what actually matters for a tortoise, not height, and the preferred size is closer to 7×3.5 ft once your budget and space allow it. An open top keeps airflow honest and cuts down on the fogging and stagnant air that a lidded glass tank traps.



NEVER

Rely on a glass-walled tank as the full-time home. Poor top ventilation lets humidity and stale air build up, and the walls encourage pacing along reflections instead of using the floor space the way a tortoise actually needs to.

What the diagram doesn't show

- Outdoor pens are a strong warm-season option once conditions allow it. Wall depth needs to run at least 12 inches below grade, since Russian tortoises dig and will find any shortcut under a shallow wall.
- Substrate depth matters as much as footprint: 6 to 12 or more inches of a 50/50 topsoil and coconut coir mix supports real digging and burrowing behavior, not just moisture retention.
- Keep the tortoise table (or enclosure) away from cold drafts and out of a high-traffic path, even though the open top makes airflow easier to manage than a sealed tank.

Temperature, UVB & humidity

ZONE	TARGET
Basking surface	95 to 100°F
Cool zone	70s°F
Nighttime	Around 60°F
Humidity	30 to 50%

Measure basking surface temperature with an infrared thermometer aimed at the actual substrate or platform, not the air above it. Confirm cool-zone and nighttime readings the same way rather than guessing by feel.

Two fixtures, not one

UVB and heat are two separate jobs. A T5 HO UVB tube provides UV but little usable heat on its own, so pair it with a separate basking bulb over the same platform to reach the 95 to 100°F target, and confirm the surface temperature with an infrared thermometer rather than trusting the bulb's wattage.

UVB is not optional

Use a linear T5 HO tube (Arcadia 12% or Zoo Med ReptiSun 10.0 are commonly recommended), positioned over the basking end so it covers roughly a third to half of the enclosure. Never place it behind glass or plastic, both block UVB almost entirely. This fixture is for the indoor setup only: outdoors, natural sunlight is a superior UVB source and no fixture is needed.

UVI targets

Aim for a UV Index of **3.0 to 4.0** at the basking area, with a gradient down toward the cool zone so your tortoise can choose its own exposure level.

Bulb tracking

Linear fluorescent UVB output drops well before the bulb visibly dims. Track the install date on the Bulb & Equipment Log on page 19 and replace it on a schedule, not by appearance.

NICE TO HAVE

A bright daylight light around 6500K alongside the UVB tube supports natural activity and grazing behavior through the day.

Signs your setup needs adjusting

YOU NOTICE	LIKELY MEANS
Substrate staying damp, glass or plastic panels fogging	Humidity running too high for the space
Tortoise stays buried or inactive, rarely basks	Basking area too cool, or fixture not covering it well
Tortoise digs constantly at the coolest corner	Enclosure running too hot overall
Tortoise actively avoids the basking area	UVI too strong for the space, or surface too hot

Brumation & seasonal care

Russian tortoises are physiologically programmed to brumate: a period of dormancy lasting roughly 2 to 4 months over winter at cooler temperatures, around 40 to 55°F. In the wild this is not optional. In an indoor home, it is, and that's a genuine decision worth thinking through rather than defaulting into.

If you choose to brumate

A pre-brumation vet check is worth doing first, to confirm your tortoise is a healthy weight and free of parasites or illness before several months of reduced activity. Expect roughly 2 to 4 months of dormancy at 40 to 55°F, then a return to full basking temperatures, UVB, and feeding once it's over.

If you choose to stay active

Keep full basking heat, UVB, and normal feeding running straight through winter. A tortoise kept this way has no physiological need to slow down and can stay active and eating year-round. Neither choice is more correct than the other.

A GENUINE CHOICE

Many tortoise species don't offer this option cleanly. Russian tortoises can go either way as long as you commit fully: either maintain full heat, light, and feeding all winter, or let a properly cooled, vet-checked brumation run its course. Half measures, like letting temperatures drift down without the deliberate 40 to 55°F setup, are the outcome to avoid.

Warm-season outdoor time

When weather allows, supervised outdoor time is a genuine benefit: natural sunlight, varied terrain, and room to move all support the same basking and activity needs the indoor setup is built around. An outdoor pen needs walls at least 12 inches deep below grade, since a tortoise that wants out will dig rather than climb.

Diet & feeding

Russian tortoises are herbivores, and weeds and leafy greens should make up the large majority of every meal. This is not a species where insects, meat, or high-protein foods belong on the regular menu.

CATEGORY	ROLE	NOTES
Weeds & leafy greens	Staple, bulk of every meal	Dandelion, plantain, clover, and similar forage; rotate varieties daily or every other day
Fruit	Occasional treat only	Under 5% of total intake if given at all
Calcium supplement	Shell & bone support	Dust forage regularly as part of the normal feeding routine

Why rotation matters

Offering fresh forage and rotating the mix of weeds and greens daily or every other day keeps the diet varied and closer to what a tortoise would encounter grazing naturally, rather than settling on one or two convenient items long-term.

NEVER FEED

Beans, peas, or any high-protein food, dog or cat food, avocado, or anything sprayed with pesticide or fertilizer. A diet that runs too rich in protein and calcium relative to fiber is a leading driver of pyramiding, see page 11.

Hydration

Offer a shallow water dish at all times, and a regular soak supports hydration alongside the moisture already in fresh forage. Adequate hydration and activity also play a role in preventing pyramiding, alongside diet, see page 11.

Common mistakes & enrichment

Mistakes that cause the most harm

- Undersizing the enclosure "because it's small for now," then never upgrading toward the preferred 7×3.5 ft footprint.
- Skipping UVB entirely, or using a coil or compact bulb instead of a proper linear T5 HO tube.
- Feeding a protein- or fruit-heavy diet instead of weeds and greens, a leading cause of pyramiding.
- Keeping substrate too shallow to allow real digging and burrowing behavior.
- Letting winter temperatures drift down by accident instead of making a deliberate brumation decision (page 7).
- Keeping substrate too damp and dirty, which invites shell rot (page 12).
- Buying a cheap wild-caught adult without weighing the welfare and conservation cost against a captive-bred tortoise, see page 15.

Enrichment, by priority

Foraging

Scatter-feeding and rotating forage daily or every other day engages natural grazing behavior instead of routine bowl-feeding.

Digging area

Full-depth substrate (6 to 12+ in) lets a tortoise dig and burrow the way it would in the wild, not just walk on top of it.

Varied terrain

Rocks, logs, and gentle slopes encourage more natural movement and grazing paths than a flat, empty floor.

Outdoor time

Supervised outdoor access in warm weather, in a dig-proof pen, adds natural sunlight and terrain variety indoor setups can't fully match.

Health red flags

What to watch for, and what to actually tell the vet.

CONTACT A VET PROMPTLY IF YOU SEE

Refusal to eat outside of a planned brumation, sunken eyes, discharge from eyes/nose/mouth, open-mouth breathing, soft or swollen limbs, lethargy with no clear trigger, straining without producing waste, or any soft, discolored, or foul-smelling patch on the shell.

What to tell the vet

- ☐ Current basking & cool-zone temps, and UVB bulb age
- ☐ Whether brumation is planned, underway, or not happening this year
- ☐ Diet breakdown over the last two weeks
- ☐ Any recent change in shell texture, color, or shape

Why UVB and diet come up first

Most of the conditions in this section, metabolic bone disease, pyramiding, and shell rot, trace back to a UVB, calcium, diet, or substrate problem more often than anything else. A vet will ask about your setup before anything else for exactly that reason.

Normal vs. concerning

NORMAL	CONCERNING
Reduced appetite and activity during a vet-cleared, deliberate brumation period	Loss of appetite or lethargy outside of brumation, or dormancy running well beyond the typical 2 to 4 month window
Occasional light digging or shallow burrowing in substrate	Persistent straining without producing waste
Brief retreat into the shell when startled or handled	Staying withdrawn for extended periods, paired with swelling or discharge

Metabolic bone disease & pyramiding

Metabolic Bone Disease (MBD)

Cause: insufficient UVB or calcium relative to phosphorus over time, the same underlying mechanism seen across UVB-dependent reptiles. **Signs:** soft or misshapen shell, difficulty walking, tremors, general weakness.

Diagnosis: physical exam plus an X-ray to check bone and shell density. **Response:** vet diagnosis and treatment promptly, this does not self-correct. An illness workup like this commonly runs \$150 to \$300 or more.

Recovery: mild, early cases can improve over weeks to months with corrected UVB and vet-guided calcium; advanced shell or bone deformity often persists even once the disease itself is controlled.

PREVENTION BEATS TREATMENT

Correct UVB (page 6) and a diet built around weeds and greens with regular calcium (page 8) prevent nearly all MBD cases. This is the single highest-leverage thing in this whole package.

Pyramiding

Cause: a diet too rich in protein or calcium relative to fiber, or insufficient humidity and activity. **Signs:** abnormal, bumpy, raised growth on the shell scutes instead of smooth, even growth. **Diagnosis:** visual and physical exam.

REALITY CHECK

Pyramiding cannot be reversed once it happens. This is a prevention-only issue: the response is correcting diet, humidity, and activity going forward, not undoing the existing shell shape. That's exactly why the diet guidance on page 8 matters so much.

Respiratory infection, shell rot & parasites

Three conditions worth knowing by name before you ever see them.

Respiratory Infection

Cause: most often temperatures running too cool for too long. **Signs:** open-mouth breathing, discharge from the nose or mouth, lethargy. **Diagnosis:** physical exam and listening to the lungs, the vet will ask about your temperature setup directly. **Response:** vet visit for treatment; correct the basking and cool-zone temperatures in parallel or it recurs.

Shell Rot

Cause: usually a too-damp, dirty substrate held against the shell over time. **Signs:** soft, discolored, or foul-smelling patches on the carapace or plastron. **Diagnosis:** physical exam. **Response:** vet treatment, and correcting substrate moisture and cleanliness so it doesn't return.

Parasites

Diagnosis: a fecal exam. **Reality check:** internal parasites are common in tortoises generally, especially wild-caught individuals or ones recently acquired from an unknown history. **Response:** vet-guided treatment, followed by a fecal recheck.

THE COMMON THREAD

All three trace back to setup or sourcing: temperature for respiratory infection, substrate moisture and cleanliness for shell rot, and unknown history for parasites. A captive-bred tortoise from a known source, kept at correct temperatures on clean, appropriately moist substrate, avoids most of this.

Setup checklist & temperature/UVB targets

Essential setup checklist

- | | |
|--|---|
| ☐ 4×2×2 ft open-top tortoise table minimum (7×3.5 ft preferred) | ☐ 50/50 topsoil & coconut coir substrate, 6 to 12+ in deep |
| ☐ Linear T5 HO UVB tube over the basking end | ☐ Shallow water dish |
| ☐ Bright 6500K daylight light alongside the UVB | ☐ Hide options at both warm and cool ends |
| ☐ Basking area reaching 95 to 100°F | ☐ Weeds/greens forage source lined up |
| ☐ Infrared thermometer | ☐ UVB bulb install date tracked |

Temperature & UVB targets

TARGET	RANGE
Basking area	95 to 100°F
Cool zone	70s°F
Nighttime	Around 60°F
Humidity	30 to 50%
UVI at basking area	3.0 to 4.0
Brumation window (if brumating)	2 to 4 months at 40 to 55°F

Before your tortoise comes home

- | | |
|--|---|
| ☐ Enclosure fully set up and running for 24 to 48 hours first | ☐ UVB fixture mounted and covering the basking end as intended |
| ☐ Basking and cool-zone temps confirmed with a thermometer, not by feel | ☐ Weeds and greens source lined up before day one |

Emergency & quick targets

CALL A REPTILE VET NOW IF

Open-mouth breathing or discharge · soft, discolored, or foul-smelling shell patches · sunken eyes with lethargy · straining without producing waste · refusal to eat outside of a planned brumation · limb swelling or tremors

READING	TARGET
Basking area	95 to 100°F
Cool zone	70s°F
Nighttime low	Around 60°F
Humidity	30 to 50%
UVI at basking area	3.0 to 4.0
Brumation window (if brumating)	2 to 4 months at 40 to 55°F

HAVE THIS READY WHEN YOU CALL

Basking & cool-zone temps, UVB bulb age, diet over the last two weeks, whether brumation is happening this year, and any recent shell changes.

Budget & shopping list

ITEM	TYPICAL COST
Tortoise (hatchling, captive-bred)	\$50 to \$150
Tortoise (older, captive-bred juvenile/adult)	\$225 to \$400
Complete indoor setup (table, UVB, heat, substrate, furnishings)	\$400 to \$600
of which, the open-topped tortoise table alone	\$125 to \$200
Initial vet check	Around \$75
Annual ongoing cost	\$500 to \$650

PLAN FOR, DON'T COUNT ON AVOIDING

An illness workup commonly runs \$150 to \$300 or more. A small emergency fund matters as much as any item on this list.

WHERE NOT TO CUT CORNERS

The UVB fixture and the size of the enclosure are the two line items most directly tied to the health conditions in this package. Save on decor and extras before you save here.

A CHEAP TORTOISE ISN'T ALWAYS A GOOD DEAL

Wild-caught adults occasionally show up for under \$50, undercutting captive-bred prices, but wild collection pressure is part of why this species is listed Vulnerable, and wild-caught individuals carry a higher parasite and disease burden (page 12). Captive-bred is the better choice even at a higher upfront price.

First 30 days & days 31 to 90

Week 1

- ☐ Confirm all temperatures and UVI with an actual thermometer/meter, not "feels warm"
- ☐ Let your tortoise settle: minimal handling, focus on feeding and observing appetite
- ☐ Schedule a wellness vet visit within the first couple of weeks

Week 2 to 3

- ☐ Establish a daily or every-other-day forage rotation, weeds and greens dominant
- ☐ Begin a consistent calcium supplement routine
- ☐ Start the Owner Log on page 19

Week 4

- ☐ Review weight and shell condition trends from the Owner Log
- ☐ Mark your UVB bulb's install date on the Bulb & Equipment Log
- ☐ Introduce first enrichment items from page 20

Days 31 to 90

- ☐ Book the follow-up wellness check most vets recommend after initial setup
- ☐ Re-measure UVI directly if you have access to a meter, don't rely on the bulb's age alone
- ☐ Look back at the Owner Log for any slow drift in weight, appetite, or shell condition
- ☐ If winter is approaching, work through the brumation decision guide on page 7 and book a pre-brumation vet check if you're choosing to brumate

Symptom quick reference

SYMPTOM	LIKELY CAUSE	ACTION
Reduced appetite/activity, cooler months	Possible brumation	Vet check first, then follow the decision guide (page 7)
Refusal to eat outside of brumation	Illness, temperature, or setup problem	Vet visit
Soft, discolored, or foul-smelling shell patches	Shell rot	Vet promptly (page 12)
Bumpy, raised shell growth pattern	Pyramiding	Correct diet/humidity going forward (page 11); prevention only
Soft or misshapen shell, difficulty walking	Metabolic bone disease	Vet immediately (page 11)
Open-mouth breathing, discharge	Respiratory infection	Vet promptly (page 12)
Persistent diarrhea or abnormal stool	Parasites	Vet visit, fecal exam (page 12)
Straining without producing waste	Possible blockage or illness	Vet visit
Sudden weight loss between log entries	Several possible causes	Vet visit, bring the Owner Log (page 19)
Repeated digging at the cool zone, avoiding basking area	Setup running too hot, or UVI too strong	Recheck temps and UVB (page 6)

Daily, weekly, monthly & seasonal routine

Daily

- ☐ Check basking & cool-zone temps
- ☐ Offer fresh weeds/greens forage, rotating varieties daily or every other day
- ☐ Refresh water dish
- ☐ Quick visual health check: eyes, shell, posture, appetite

Weekly

- ☐ Spot-clean substrate and wipe down water dish
- ☐ Weigh and log in the Owner Log
- ☐ Rotate or refresh one enrichment item
- ☐ Confirm UVB bulb is still within its tracked replacement window

Monthly

- ☐ Check substrate depth and top up toward 6 to 12+ inches as needed
- ☐ Review the Owner Log for slow trends, not just single readings
- ☐ Check that hides and terrain still fit your tortoise's current size

Seasonal

- ☐ Reassess indoor/outdoor time as the weather changes
- ☐ Make and follow through on the brumation decision each winter (page 7); book a pre-brumation vet check if brumating

Owner log

Photocopy this page, or track digitally. Consistency matters more than the format.

DATE	WEIGHT	APPETITE	SHELL CONDITION	BASKING BEHAVIOR	NOTES

WHY BOTHER LOGGING

A steady weight and shell-condition trend is the single best early warning system you have. Most of the conditions in this package show up as a change here before anything else is visible.

Bulb & equipment log

ITEM	INSTALLED	REPLACE BY
UVB bulb (T5 HO)		
6500K daylight bulb		
Thermometer/hygrometer batteries		

Vet visit log

DATE	REASON	FOLLOW-UP

Enrichment checklist

Setup

- ☐ Varied terrain: rocks, logs, and gentle slopes for natural grazing movement
- ☐ Full-depth digging/burrowing substrate area, 6 to 12+ inches
- ☐ Hide options at both warm and cool ends

Monthly review

- ☐ Rotate or reposition one terrain feature to refresh the layout
- ☐ Offer scatter-feeding or forage rotation at least once
- ☐ Provide supervised outdoor time when weather allows
- ☐ Reassess whether current terrain still fits your tortoise's size

Enrichment log

DATE	ACTIVITY	HOW IT WENT

Glossary

Brumation	A reptile's natural winter dormancy period. For Russian tortoises kept indoors with full heat, light, and feeding year-round, it's optional rather than required.
Pyramiding	Abnormal, bumpy shell growth instead of smooth growth, largely irreversible once it has occurred.
MBD	Metabolic Bone Disease: weakened bones and shell from inadequate UVB or calcium.
UVI	UV Index: the measurement scale used to describe basking-area UVB intensity.
Carapace	The domed upper shell.
Plastron	The flat underside shell.
Fecal exam	A lab check of a stool sample, used to diagnose internal parasites.
Tortoise table	An open-topped enclosure design that prioritizes floor space and ventilation over height.
Captive-bred	Bred in captivity rather than collected from the wild; preferred for both welfare and conservation reasons.

Compiled from veterinary and herpetological sources, cross checked against current Beastly Facts guides at time of publication. This package is a husbandry reference, not a substitute for a reptile experienced veterinarian, and is not a live website mirror. No external links or website CTAs appear in this document.

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